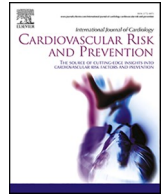




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Position paper ITACARE-P/FADOI on the referral from internal medicine to cardiac rehabilitation: Executive summary and factsheet[☆]

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ABSTRACT

The Italian Association for Cardiovascular Rehabilitation and Prevention (ITACARE-P) and the Italian Federation of Associations of Hospital Doctors on Internal Medicine (FADOI) released a joint position paper to guide referrals of cardiovascular patients discharged from Internal Medicine (IM) wards to Cardiac Rehabilitation (CR) facilities. The document provides rationale and operative recommendations for appropriateness (i.e. qualifying diagnoses) and priority criteria to overcome the mismatch between potential demand and effective supply of CR programmes. In case of no-referral due to logistic barriers, the document recommends the adoption of best alternatives to CR for disability reduction, better prognosis, and improvement of quality of life. The joint position paper is also aimed to promote the consideration of IM as a potential stakeholder of CR.

1. Rationale

Cardiac Rehabilitation (CR) is recognized as a class I intervention for clinical stabilization, risk reduction, treatment of disability, and for the improvement of quality of life across a wide range of cardiovascular (CV) conditions [1]. CR programmes may be delivered both in the residential and ambulatory setting, the first one being used in a fast-track modality by acute wards. Patients are usually referred after an index hospitalization for acute coronary syndromes (ACS), coronary revascularization (both percutaneous or surgical), heart failure (HF), and valve surgery. According to ESC Guidelines, the CR programme is based on specific core components (i.e. patient evaluation, physical activity counselling, individualized training prescription, lipid and blood

pressure management, diet/nutritional counselling, psychosocial support) and delivered by a multi-disciplinary team consisting of nurses, physiotherapists, psychologists, and dietitians [2].

Internal medicine (IM) patients are mostly elderly and with multiple complex co-morbidities [3,4], potentially being appropriate for referral to CR after hospitalization for ACS, HF, cardiac electronic device implantation, or peripheral arterial disease (PAD) [5].

By acknowledging the importance of IM as a stakeholder, the Italian Association for Cardiovascular Rehabilitation and Prevention (ITACARE-P) and the Federation of Associations of Hospital Doctors on Internal Medicine (FADOI) released a joint position paper to guide referrals of IM patients to CR programmes, published in Italian [6] and offered to the whole national community. This summary is aimed to

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Table 1
ITACARE-P and FADOI recommendations for referral priorities of Internal Medicine patients to Cardiac Rehabilitation.

TARGET POPULATION	REFERRAL PRIORITIES
Patients after acute coronary syndromes	To residential CR High risk of heart failure or progressive left ventricular dysfunction, represented by: -congestive heart failure during the index hospitalization -LVEF <40 % -LVEF between 40 % and 45 % and at least one negative predictor (i.e. moderate to severe mitral valve regurgitation, diastolic dysfunction, high left ventricular asynergy score, or long/complicated hospitalizations during the acute phase). To ambulatory CR High thrombotic/ischemic risk, represented by: -diabetes -chronic kidney disease -PAD -history of angina or previous ACS -multivessel coronary disease, complex lesions, lack or incomplete coronary revascularization, previous stent thrombosis. Patients with heart failure To residential CR After hospitalization for hemodynamic instability with high relapse risk, represented by: -NYHA III-IV class at discharge -need of inotropic or vasodilators infusion -electrical instability -clinical instability -need of drug uptitration -need of evaluation for VAD or HT -need of evaluation for CIED To ambulatory CR Chronic heart failure with high disability in absence of criteria for residential CR Patients with chronic coronary syndromes To residential CR High cardiac-related disability (e.g. angina on minimal effort or at rest; high risk of heart failure, high thrombotic risk, high risk of arrhythmias). To ambulatory CR CCS with high disability in absence of criteria for residential CR Patients with peripheral arterial disease To residential CR -after peripheral revascularization with high risk of subsequent MACE or MALE -with trophic lesions -with hemodynamic or electrical instability -with need of prostanoid infusion therapy To ambulatory CR With intermittent claudication to improve walking ability, achievement of cardiovascular prevention targets, and healthy lifestyle

ACS: acute coronary syndrome; CCS: chronic coronary syndromes; CIED: cardiac implantable electronic device; CR: cardiac rehabilitation; HT: heart transplantation; LVEF: left ventricular ejection fraction; MACE: major adverse cardiovascular event; MALE: major adverse limb event; NYHA: New York Heart Association; PAD: peripheral arterial disease; Ventricular Assistance Device.

provide essential information on operational steps, in the form of basic questions. Other details on epidemiology, risk stratification, phenotypic features, CR national network, and current legislation can be found in the source publication.

2. Why to refer from the IM to the CR setting

- Because CR is a comprehensive and cost-effective secondary prevention program consisting of clinical stabilization, pharmacological optimization, supervised exercise training, nutritional intervention, psychosocial support, and lifestyle education, thus covering several needs of CV patients discharged from IM wards.

- Because CV patients admitted to IM often display worse clinical situation and higher CV risk (e.g. due to absence of coronary revascularization) as compared to those admitted to cardiology units.
- Because the CR setting offers multidisciplinary approach and expertise for the management of conditions often associated to IM patients, such as frailty, sarcopenia, multimorbidity, and polytherapy.
- Because - despite clear benefits - the use of CR remains low, with only 6 % of patients referred to a CR programme (both residential or ambulatory) after ACS in Italy in 2021 [7].

3. When to refer

- To residential (in-patient) programmes: immediately after or within 15 days (fast-track modality) after hospital discharge following a qualifying cardiac event.
- To ambulatory (out-patients) programmes: in the following 12 months after discharge from IM with a qualifying cardiac event and no indication/priority for residential CR intervention due to clinical stability.

4. Who are eligible patients

All patient populations as defined in current guidelines [1], i.e.:

- Post ACS
- Post primary and elective angioplasty
- Chronic coronary syndromes (CCS)
- Coronary artery surgery
- Valve surgery (including procedures such as TAVI and MitraClip)
- Chronic or recent exacerbation of HF
- Cardiac transplantation
- Cardiac Implantable Electronic Device (CIED) implantation
- Peripheral Arterial Disease (PAD)

Among referral groups, patients with diabetes, chronic heart failure, and cancer represent challenging populations with dedicated rehabilitative programs.

Beware of ensuring equities and similar opportunities to selected disadvantaged populations, such as women, elder patients, minorities and people of lower socioeconomic status.

5. Who to prioritize

According to the last available survey carried out by the national society of cardiology in 2022 [8], in Italy 116 CR facilities with residential programmes and 147 facilities with ambulatory programmes are currently available, with a total number of 2075 CR beds. It has been estimated that the national CR network could only accommodate less than 30 % of all CV eligible patients. Given the mismatch between potential demand and effective supply of CR programmes, priority criteria were elaborated after ACS [9], reproduced here with specific adaptation to the IM setting and now including additional clinical conditions such as CCS, HF, and PAD (Table 1).

6. How to refer

- Check for patient’s appropriateness for referral based on CV index event.
- Establish goals for clinical improvements, functional recovery, and lifestyle modifications in accordance with expected patient adherence and life expectancy.
- Select the CR facility by checking declared structure- and process-based metrics.
- Establish formal relationship between the IM and the CR facility, also promoting automatic referral systems.

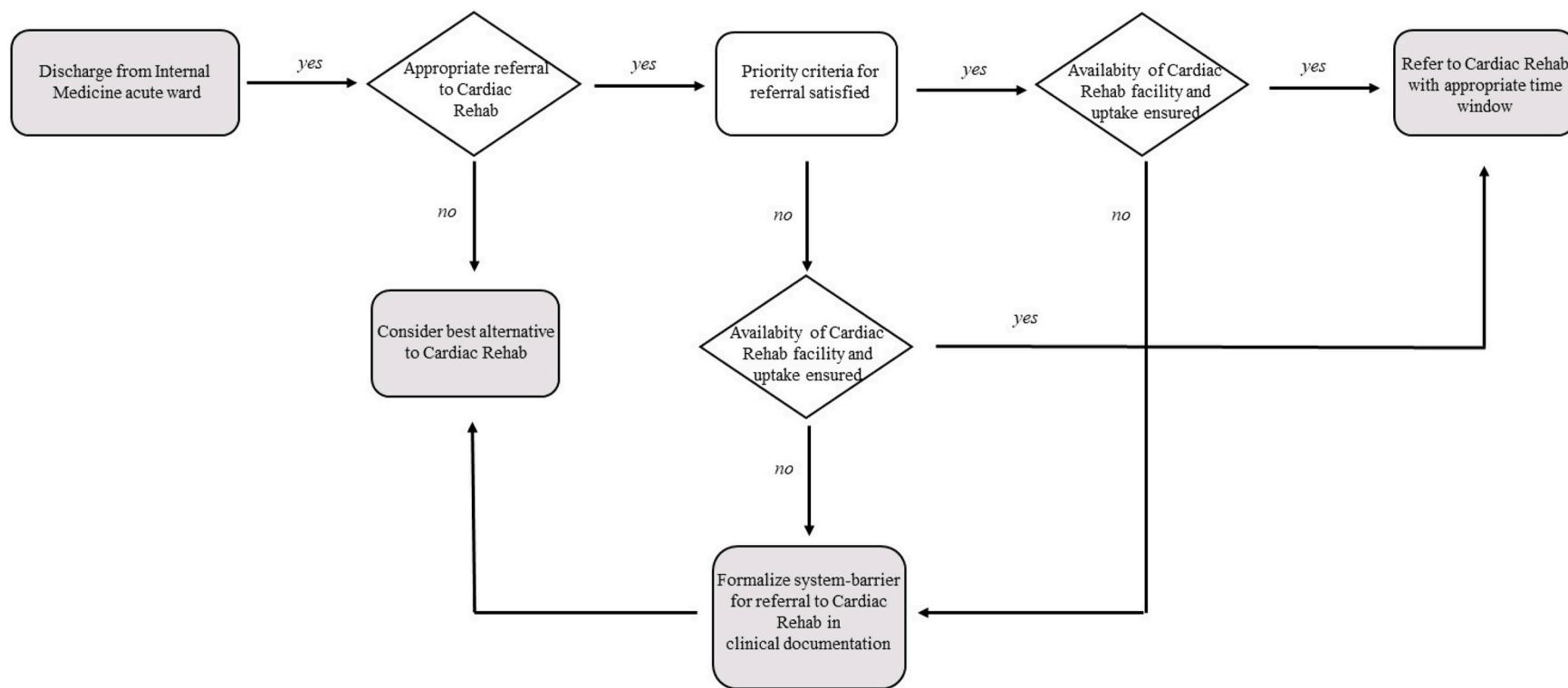


Fig. 1. Flow-chart for referral from Internal Medicine to Cardiac Rehabilitation.

- Explore patient- and system-related factors limiting enrolment and completion rates of CR, especially for ambulatory programs (e.g. low socioeconomic status, low health literacy, lack of perceived need of CR, work-related factors, absence of caregiver, distance from home).
- Proceed to refer in accordance with local legislation (e.g. the new policies of the Italian Health System [10]).

7. What to do in case of no-referral

When referral to CR is not allowed in eligible and priority patients due to logistic barriers (e.g. lack of CR facility), the panel recommends specifying the reason in clinical documentation and to check for best alternatives (Fig. 1). Alternatives to centre-based (both residential or ambulatory) programmes may be found within the CR setting in form of community, telerehabilitation, or digital health tool-based interventions, if available. Most frequently, they could be represented by other residential settings offering “general rehabilitation” or physiotherapy in tertiary care. All these alternative models should be checked for real potential of disability reduction and improvement of prognosis and quality of life.

8. Conclusion

The ITACARE-P and FADOI joint position paper provides practical recommendations for the referral to CR of CV patients discharged from IM facilities, mainly focused on the Italian epidemiological situation and organization of the health system. Main drivers of the referral process are the identification of qualifying diagnoses and priorities, as far as of best alternatives in case of obliged no-referral. Above all, the position paper reaffirms the need to look to IM for a modern referral also, thus limiting the proportion of patients who do not benefit from CR.

Disclaimer statement

The ITACARE-P proceedings in this supplement were reviewed and

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